

CLASS-8

LESSON-9 MENSURATION

(This PDF Based on NCERT Book)

AREA OF A POLYGON()-

1. Triangle (3 Sides)

The most basic polygon. The formula depends on what information you have.

- **General Formula:**

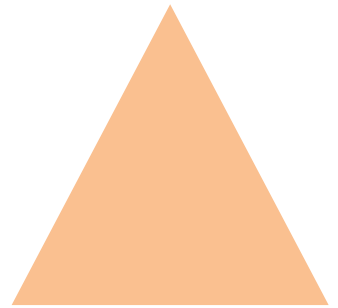
$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

- **Equilateral Triangle (Regular):**

$$\text{Area} = \frac{\sqrt{3}}{4} \times s^2$$

Example: If an equilateral triangle has a side of **4 cm**:

$$\text{Area} = \frac{\sqrt{3}}{4} \times 4^2 = \frac{\sqrt{3}}{4} \times 16 = 4\sqrt{3} \approx 6.93 \text{ cm}^2$$



2. Quadrilaterals (4 Sides)

These formulas vary based on the properties of the four-sided shape.

Shape	Area Formula	Variables
Square	S^2	s = side
Rectangle	$l \times w$	l = length, w = width
Parallelogram	$b \times h$	b = base, h = vertical height
Rhombus	$\frac{d_1 \times d_2}{2}$	d_1, d_2 = lengths of diagonals

Shape	Area Formula	Variables
Trapezoid	$\frac{a+b}{2} \times h$	a, b = parallel sides, h = height

Example (Trapezoid): A trapezoid has parallel bases of **8 cm** and **12 cm**, and a height of **5 cm**.

$$\text{Area} = \frac{8+12}{2} \times 5 = \frac{20}{2} \times 5 = 10 \times 5 = 50 \text{ cm}^2$$

3. Regular Hexagon (6 Sides)

Hexagons are common in geometry because they are made of six equilateral triangles.

- **Formula:**

$$\text{Area} = \frac{3\sqrt{3}}{2} \times s^2$$

Example: A regular hexagon with a side of **2 cm**.

$$\text{Area} = \frac{3\sqrt{3}}{2} \times 2^2 = \frac{3\sqrt{3}}{2} \times 4 = 6\sqrt{3} \approx 10.39 \text{ cm}^2$$

4. Regular Octagon (8 Sides)

Often seen in "Stop" signs.

- **Formula:**

$$\text{Area} = 2(1 + \sqrt{2}) \times s^2$$

Example: A regular octagon with a side of **3 cm**.

$$\text{Area} = 2(1 + 1.414) \times 3^2 = 2(2.414) \times 9 = 4.828 \times 9 \approx 43.45 \text{ cm}^2$$

SOLID SHAPES()

Solid shapes (also known as **3D shapes**) are objects that take up space. Unlike flat shapes (2D) like a square or a circle, which only have length and width, solid shapes have three dimensions: **length, width, and height**.

Think of a flat circle drawn on paper versus a basketball you can hold—the basketball is a solid shape.

Key Parts of a Solid Shape

To understand these shapes, we look at three main features:

1. **Face:** The flat or curved surface of the shape.
2. **Edge:** The line where two faces meet.
3. **Vertex (Corner):** The point where edges meet.

Common Solid Shapes

1. Cube

A cube is like a box where every side is a perfect square. It has 6 faces, 12 edges, and 8 vertices.

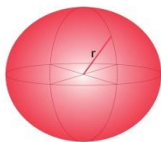
- **Real-life example:** A die (from a board game) or a Rubik's cube.

2. Sphere

A sphere is perfectly round, like a ball. It has no flat faces, no edges, and no corners. It only has one curved surface.

- **Real-life example:** A football, a marble, or the Earth.

Sphere

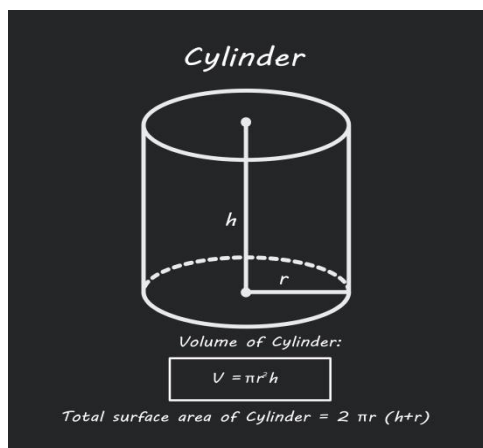


$$V = \frac{4}{3} \pi r^3$$

3. Cylinder

A cylinder has two flat circular ends and one curved side connecting them. It's shaped like a tube.

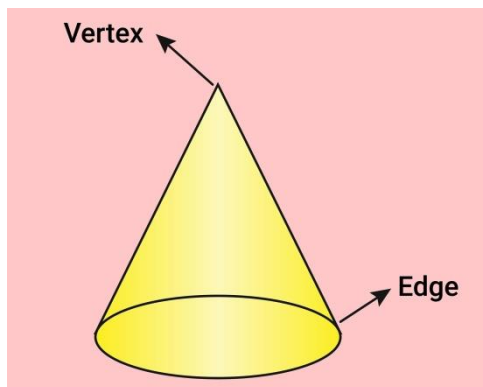
- **Real-life example:** A soda can, a battery, or a glue stick.



4. Cone

A cone has a flat circular base that narrows down to a single point at the top.

- **Real-life example:** An ice cream cone or a party hat.

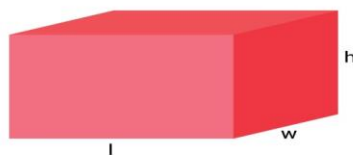


5. Rectangular Prism (Cuboid)

This is like a cube but stretched out. Its faces are rectangles.

- **Real-life example:** A brick, a cereal box, or a textbook.

Rectangular Prism



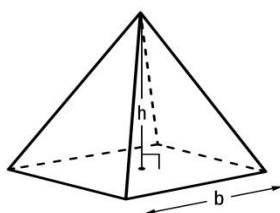
$$l \times w \times h$$

6. Pyramid

A pyramid has a flat base (usually a square) and triangular sides that meet at a point at the top.

- **Real-life example:** The Great Pyramids of Egypt or a tent.

Volume of a Square Pyramid



$$\text{Volume (V)} = \frac{1}{3} b^2 h$$

b = base, h = height

Summary Table

Shape	Faces	Edges	Vertices
Cube	6	12	8
Sphere	1 (curved)	0	0
Cylinder	3 (2 flat, 1 curved)	2	0
Cone	2 (1 flat, 1 curved)	1	1
Rectangular Prism	6	12	8

SURFACE AREA OF A CUBE, CUBOID AND CYLINDER()-

Surface area is the total area of all the outside faces of a 3D object. Imagine you are wrapping these shapes in paper; the amount of paper needed to cover them completely is the surface area.

1. Cube

A cube has 6 identical square faces. Since the area of one square is side $\times$ side (s^2), we simply multiply that by 6.

- **Formula:**

$$\text{Total Surface Area} = 6s^2$$

- **Variable:** s = length of a side (edge)

Example:

If you have a Rubik's cube where each side is **5 cm**:

- $\text{Area} = 6 \times 5^2$
- $\text{Area} = 6 \times 25$
- **Total Surface Area = 150 cm²**

2. Cuboid (Rectangular Prism)

A cuboid has 6 rectangular faces. These faces come in 3 matching pairs (top/bottom, front/back, and left/right).

- **Formula:**

$$\text{Total Surface Area} = 2(lb + bh + hl)$$

- **Variables:** l = length, b = breadth (width), h = height

Example:

Imagine a shoebox with length **10 cm**, breadth **6 cm**, and height **4 cm**:

- $\text{Area} = 2 \times (10 \times 6 + 6 \times 4 + 4 \times 10)$
- $\text{Area} = 2 \times (60 + 24 + 40)$
- $\text{Area} = 2 \times (124)$
- **Total Surface Area = 248 cm²**

3. Cylinder

A cylinder has two circular bases and one curved side.

- **Curved Surface Area (CSA):** $2\pi rh$ (just the "tube" part)
- **Total Surface Area (TSA):**

$$2\pi r(r + h)$$

- **Variables:** r = radius of the base}, h = height of the cylinder}, $\pi \approx 3.14$

Example:

A soda can has a radius of **3 cm** and a height of **10 cm**:

- $\text{Area} = 2 \times 3.14 \times 3 \times (3 + 10)$
- $\text{Area} = 18.84 \times 13$
- **Total Surface Area $\approx$ 244.92 cm²**

Quick Comparison Table

Shape	Formula	Key Feature
Cube	$6s^2$	All sides are equal
Cuboid	$2(lb + bh + hl)$	Opposite faces are equal
Cylinder	$2\pi r(r + h)$	Includes 2 circles and 1 curved side